

Minimal sequence detection algorithm on asynchronous spiking convolution neural network

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1 Introduction

2 Problem formalization

2.1 Integrate and fire neuron

Let's take the simplest spiking neuron model. The membrane potential is initialized and also reset after any spike to 0, it cannot decrease below a minimum value $\underline{v} < 0$, and if it exceeds a firing threshold $\bar{v}$ the neuron will spike.

The most important assumption is that the neuron model is pure integrate-and-fire without any leakage, because of two reasons:

1. Only relative order but not absolute timing matters in the problem.
2. There is no efficient way to implement leakage with pure asynchronous circuits.

2.2 Spiking convolution neural network

The network contains L layers, denoted as layer $l = 1, 2, \dots, L$, while layer $l = 0$ represents the input. For simplicity, assume the network is infinite and one-dimensional. The network runs purely asynchronously. For hardware implementation, consider the membrane potential and weights are all quantize to integers.

Assume layer l has C_l channels, denoted as channel $c_l = 1, 2, \dots, C_l$. Assume there is only one input channel $C_0 = 1$.

Specifically, the membrane potentials for each neuron i of layer l channel c at time t is

$$v : \mathbb{N}^2 \times \mathbb{Z} \times \mathbb{R} \mapsto \mathbb{Z}, (l, c_l, i, t) \mapsto v_i^{l,c}(t), \quad (1)$$

and the spiking status

$$s : \mathbb{N}^2 \times \mathbb{Z} \times \mathbb{R} \mapsto \{0, 1\}, (l, c, i, t) \mapsto s_i^{l,c}(t). \quad (2)$$

Assuming ideal asynchrony, simultaneous spikes are not allowed

$$\forall t \in \mathbb{R}, \sum_{l=1}^L \sum_{c=1}^{C_L} \sum_{i=-\infty}^{\infty} s_i^{l,c}(t) \leq 1 \text{ a.s.} \quad (3)$$

Feed-forward multi-casting is allowed, so the kernel weights from neuron $i - j$ ($i, j \in \mathbb{Z}$) of layer k channel c_k to neuron i of layer l channel c_l (satisfying $0 \leq k < l \leq L$) is

$$w : \mathbb{N}^4 \times \mathbb{Z} \mapsto \mathbb{Z}, (k, c_k, l, c_l, j) \mapsto w_{k,c_k}^{l,c_l}(j). \quad (4)$$

For more realistic hardware implementation, the kernel weight w should have limited fan-in or fan-out, and v and w should have finite bit-precision.

The weight w should also have “finite support”, meaning that it satisfies $\forall 0 \leq k < l \leq L, \exists J_k^l \in \mathbb{N}, \forall 1 \leq c_k \leq C_k, \forall 1 \leq c_l \leq C_l$:

$$\forall j > J_k^l, w_{k,c_k}^{l,c_l}(j) \equiv w_{k,c_k}^{l,c_l}(J_k^l) \leq 0 \wedge \forall j < -J_k^l, w_{k,c_k}^{l,c_l}(j) \equiv w_{k,c_k}^{l,c_l}(-J_k^l) \leq 0. \quad (5)$$

With these assumptions, w can be expressed “finitely” without loss of information, define shorthand

$$W_k^{l,c_l} \in \mathbb{Z}^{C_k \times (2J_k^l + 1)} : \left(w_k^{l,c_l} \right)_{1 \leq c_k \leq C_k, -J_k^l \leq j \leq J_k^l} = w_{k,c_k}^{l,c_l}(j). \quad (6)$$

When writing them out, “...” symbols could be used at the heads and tails to indicate that the values will “continue” infinitely in both directions.

If the layer has only one channel, for example, the input layer $l = 0$. the channel ID can be omitted, thus shorthand like

$$s_i^0(t) := s_i^{0,1}(t), \quad (7)$$

and

$$w_0^{l,c_l}(j) := w_{0,1}^{l,c_l}(j). \quad (8)$$

can be used.

2.3 Relative order assumptions in asynchronous network

The neurons are all integrate-and-fire without leak, so all “state” (v and s) changes are triggered only by inputs. In a continuous-time real-world physical system, the convolution operation and spike generation and propagation take both some deterministic order and some nondeterministic delay.

Formally, if source layer k is multi-cast to N target layers $\mathcal{L}_k = \{l_n\}_{n=1,\dots,N}$. If a neuron in layer k channel c_k fires at time t_{source} , layer $l_n \in \mathcal{L}_k$ will receive the spike after some propagation delay at time $t_{\text{source}} + \Delta t_{k,c_k,l_n}$ where

$$\Delta t_{k,c_k,l_n}^{\text{pd}} = \sum_{\nu=1}^n \varepsilon_{k,c_k,\nu}^{\text{pd}}, \quad (9)$$

where $\varepsilon_{k,c_k,\nu}^{\text{pd}}$ are random variables taking positive values in very small range $(0, \varepsilon_{\text{max}}^{\text{pd}})$, which guarantees that

$$0 < \Delta t_{k,c_k,l_1}^{\text{pd}} < \Delta t_{k,c_k,l_2}^{\text{pd}} < \dots < \Delta t_{k,c_k,l_N}^{\text{pd}}. \quad (10)$$

When the spike arrives in layer l_n (at time t_{target}), the “integrate” operation is done for each channel, each neuron with a single processing engine as a “nested for-loop”. For the most general case, there should not be any “preferred” spatial direction, but only on the channel dimension. namely for any spike triggered at neuron i in channel c in this layer (l_n), it is generated at time $t_{\text{target}} + \Delta t_{l_n,c,i}^{\text{pe}}$ where

$$\Delta t_{l_n,c,i}^{\text{pe}} = \sum_{\varsigma=1}^c \varepsilon_{l_n,\varsigma}^{\text{ch}} + \varepsilon_{l_n,c,i}^{\text{nrn}} \quad (11)$$

where $\varepsilon_{l_n, \varsigma}^{\text{ch}}$ are random variables taking positive values in very small range $(\varepsilon_{\min}^{\text{ch}}, \varepsilon_{\max}^{\text{ch}})$ and $\varepsilon_{l_n, c, i}^{\text{nrn}}$ are random variables taking positive values in $(0, \varepsilon_{\max}^{\text{nrn}})$ where

$$\varepsilon_{\max}^{\text{nrn}} + L\varepsilon_{\max}^{\text{pd}} < \varepsilon_{\min}^{\text{ch}} \quad (12)$$

which guarantees that for any target layer $l \in \mathcal{L}_{l_n}$, any spike from layer l_n with lower channel ID always arrives at layer l before any spike with higher channel ID.

$$\forall i_1, \dots, i_{C_{l_n}} \in \mathbb{Z}, 0 < \Delta t_{l_n, 1, i_1}^{\text{pe}} + \Delta t_{l_n, 1, l}^{\text{pd}} < \dots < \Delta t_{l_n, C_{l_n}, i_{C_{l_n}}}^{\text{pe}} + \Delta t_{l_n, C_{l_n}, l}^{\text{pd}}. \quad (13)$$

Since the network can only be triggered by external input, given an input spike at time t_0 , the network states will settled down within time $t_0 + T_{\max}$ where

$$\begin{aligned} T_{\max} &= \sum_{l=1}^L ((L-l+1)\varepsilon_{\max}^{\text{ch}} + C_l\varepsilon_{\max}^{\text{pd}} + \varepsilon_{\max}^{\text{nrn}}) \\ &= \frac{L(L+1)}{2}\varepsilon_{\max}^{\text{pd}} + \sum_{l=1}^L C_l\varepsilon_{\max}^{\text{ch}} + L\varepsilon_{\max}^{\text{nrn}} \end{aligned} \quad (14)$$

Assume that $T_{\max}$ is small enough so that there is no input spike during interval $(t_0, t_0 + T_{\max})$.

Notice that these bounds not only simplify the derivations, but also have physical meanings in asynchronous circuit design, making the analysis realistic and the algorithm implementable.

Formally, all the above-mentioned random variables can be defined on probability space $(\Omega, \mathcal{F}, \mathcal{P})$, satisfies the condition that any σ -algebra $A \in \mathcal{F}$ (which is a subset of some high-dimensional Euclidean space) with nonzero Lebesgue measure also has nonzero probability measure $\mathcal{P}(A) > 0$. This is required to formalize that the solution must work almost surely despite the stochastic timing characteristics.

2.4 Problem statement in finite state machine equivalence

This paper discusses about a way to detect a sequence of input using the above defined network model. Given that the network is a one-dimensional convolution network, without loss of generality, the problem is formulated as follows:

Given a target sequence length M , Does there exist a network which fulfills the conditions above, i.e., with L layers with $C_1, C_2, \dots, C_L$ channels, weight w , where the output layer has two layers ($C_L = 2$) that respond to inputs with consecutively increasing and decreasing IDs at the corresponding location?

For example, if $M = 4$ and the input spikes are $s_{x-3}^0, s_{x-2}^0, s_{x-1}^0, s_x^0$ ($x \in \mathbb{Z}$), neuron x of layer L channel 1 should be triggered $s_x^{L,1}$ after input spike s_x^0 , while if the input spikes are $s_{x+3}^0, s_{x+2}^0, s_{x+1}^0, s_x^0$, the output spike should be $s_x^{L,2}$. Multiple spikes from the same input should be allowed as long as the

order is maintained. The sequence will always be counted since the appearance of the first element in the encoded sequence.

The sequence to be detected could be expressed with equivalent finite state machine as shown in Fig. 1 and Fig. 2.

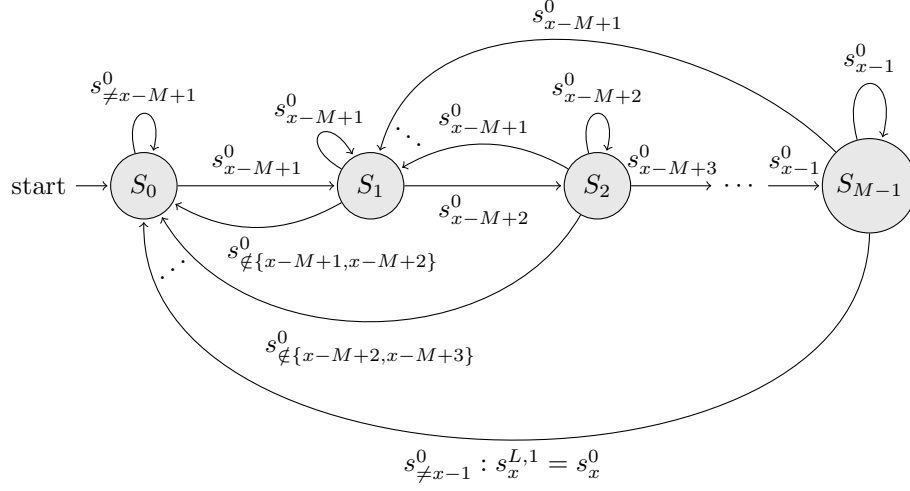


Figure 1: Equivalent finite state machine of the increasing sequence

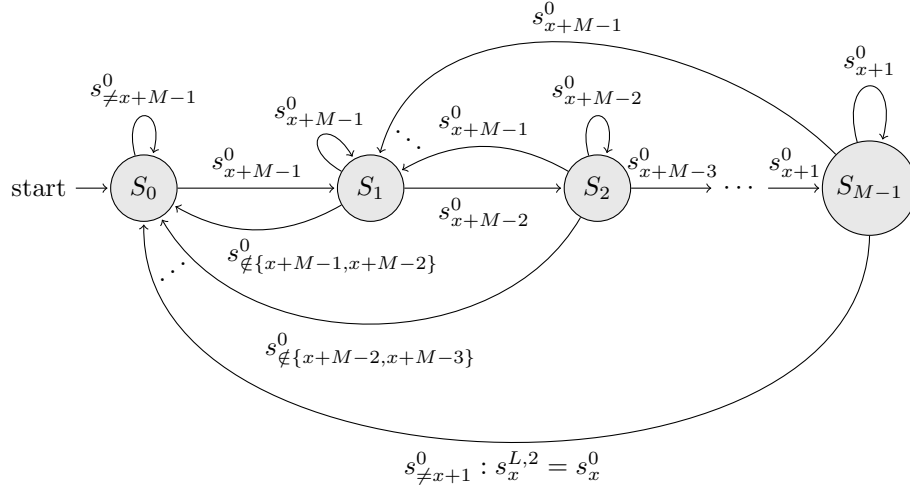


Figure 2: Equivalent finite state machine of the decreasing sequence

3 Sequence of two inputs with two layers

3.1 Infeasible with single layer

Theorem 1. *For the network with a single layer, it is impossible to encode for the sequence of two.*

Proof. By definition, the layer 1 has two channels $C_1 = 2$ encoding increasing and decreasing sequence respectively.

Consider sequence $x - 1, x$. the neuron x of channel 1 should fire exactly after input x , thus

$$\max(w_{0,1}^{1,1}(1), \underline{v}_1) < \bar{v}_1 \Rightarrow w_{0,1}^{1,1}(1) < \bar{v}_1 \quad (15)$$

$$\max(w_{0,1}^{1,1}(1), \underline{v}_1) + w_{0,1}^{1,1}(0) = \max(w_{0,1}^{1,1}(1) + w_{0,1}^{1,1}(0), \underline{v}_1 + w_{0,1}^{1,1}(0)) \geq \bar{v}_1 \quad (16)$$

Consider sequence $x, x - 1$. the neuron x of channel 1 should not fire, thus

$$\max(w_{0,1}^{1,1}(0), \underline{v}_1) < \bar{v}_1 \Rightarrow w_{0,1}^{1,1}(0) < \bar{v}_1 \quad (17)$$

$$\max(w_{0,1}^{1,1}(0), \underline{v}_1) + w_{0,1}^{1,1}(1) < \bar{v}_1 \Rightarrow w_{0,1}^{1,1}(0) + w_{0,1}^{1,1}(1) < \bar{v}_1 \quad (18)$$

Plug in (16),

$$\underline{v}_1 + w_{0,1}^{1,1}(0) \geq \bar{v}_1 \Rightarrow w_{0,1}^{1,1}(0) \geq \bar{v}_1 - \underline{v}_1 > \bar{v}_1 \quad (19)$$

contradicts with (17). $\square$

3.2 Solution with two layers

For layer 1, $C_1 = 2$, $\underline{v}_1 = -1$, $\bar{v}_1 = 1$, $J_0^1 = 1$,

$$W_0^{1,1} = W_0^{1,2} = [\dots \quad 0 \quad 1 \quad 0 \quad \dots]. \quad (20)$$

For layer 2, $C_2 = 2$ (by problem definition), $\underline{v}_2 = -1$, $\bar{v}_2 = 2$, $J_1^2 = 2$,

$$W_1^{2,1} = \begin{bmatrix} \dots & -1 & -1 & 1 & -2 & -1 & \dots \\ \dots & 0 & 0 & -1 & 2 & 0 & \dots \end{bmatrix}, \quad (21)$$

$$W_1^{2,2} = \begin{bmatrix} \dots & -1 & -2 & 1 & -1 & -1 & \dots \\ \dots & 0 & 2 & -1 & 0 & 0 & \dots \end{bmatrix}. \quad (22)$$

Proof. Let t_0 be the time layer 1 receives the input spike, it will output two spikes at the same position from channel 1 and 2, which arrive at layer 2 at times t_1 and t_2 , satisfying $t_0 < t_1 < t_2$. Consider the membrane potential of neuron x of channel 0 on layer 2 at t_1^- and t_2^+ (before receiving the first spike and after processing the second spike from layer 1), the membrane potential transition shown in Fig. 3 is equivalent to the target state machine in Fig. 1 for sequence length $M = 2$. $\square$

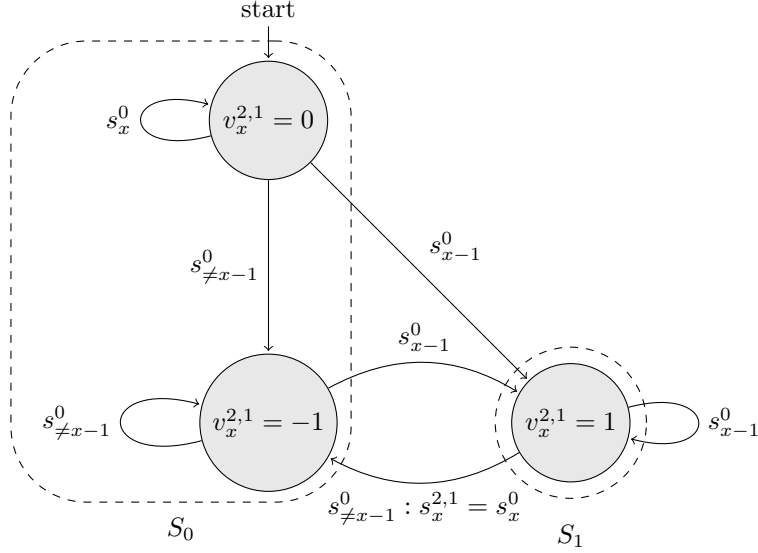


Figure 3: Membrane potential transition for detecting sequence of two

4 Sequence of three inputs with three layers

Based on solution in Sec. 3.2, the next question is how to construct a network with one more layer to detect sequence of 3.

4.1 Minimality proof

4.1.1 Infeasible with two layers

Lemma 2. *If a spike sequence changes the membrane potential of a target neuron but not causing it fire until the last spike, the change of the membrane potential can be described by a ReLU function.*

Formally, for spikes $s_1, \dots, s_m$, if $\exists t_1 < t_2 < \dots < t_m, \exists \mathcal{V} \subset [\underline{v}, \bar{v})$, starting with $v(t_1^-) \in \mathcal{V}$, after receiving the sequence

$$s_1(t_1) s_2(t_2) \dots s_m(t_m), \quad (23)$$

the following condition is satisfied

$$\forall 1 \leq i < m, v(t_i^+) < \bar{v} \quad (24)$$

where t_i^- denotes the membrane potential before receiving the corresponding spike and t_i^+ represents the membrane potential after the integration but before it generates the possible spike.

Then

$$\exists w, \exists b \geq \underline{v}, \forall v(t_1^-) \in \mathcal{V}, v(t_m^+) = \max(v(t_1^-) + w, b) \quad (25)$$

Proof. $\forall 1 \leq i < m$, the change of membrane potential $v(t_i^-) \mapsto v(t_i^+)$ is a function f_i in space

$$\mathcal{R}_{\underline{v}} := \{f(v) = \max(v + w, b)\}_{w \in \mathbb{R}, b \geq \underline{v}} \quad (26)$$

which is closed for composition operation, and since the neuron does not fire until the last spike,

$$\forall 1 \leq i < m, v(t_i^+) = v(t_{i+1}^-) \quad (27)$$

Thus

$$v(t_m^+) = \underbrace{f_m \circ \dots \circ f_1}_{\in \mathcal{R}_{\underline{v}}}(v(t_1^-)) \quad (28)$$

□

Lemma 3. $\forall f \in \mathcal{R}_{\underline{v}}$ defined in (26), $\forall v_1, v_2 \in \mathbb{R}, v_1 \leq v_2$, it is easy to prove

$$f(v_1) \leq f(v_2) \quad (29)$$

and

$$f(v_1) - v_1 \geq f(v_2) - v_2 \quad (30)$$

Lemma 4. *If a spike sequence increases the membrane potential of a target neuron but not causing it fire until the last spike, the neuron's membrane potential can be raised to the same level by repeatedly applying this sequence.*

Formally, if $\exists t_1 < t_2 < \dots < t_m, \exists v^- := v(t_1^-) \in [\underline{v}, \bar{v})$, after receiving the sequence

$$s_1(t_1) s_2(t_2) \dots s_m(t_m), \quad (31)$$

the following conditions are satisfied

$$\forall 1 \leq i < m, v(t_i^+) < \bar{v} \wedge v^+ := v(t_m^+) > v(t_1^-). \quad (32)$$

Then for all possible membrane potentials $\forall v(0) \in [\underline{v}, \bar{v})$, $\exists n \in \mathbb{N}^$, for all relative timings $\forall \{t_{ij}\}_{1 \leq \nu \leq n, 1 \leq \mu \leq m}$ satisfying*

$$0 < t_{11} < \dots < t_{1m} < t_{21} < \dots < t_{2m} < \dots < t_{n1} < \dots < t_{nm}, \quad (33)$$

after receiving sequence

$$s_1(t_{11}) \dots s_m(t_{1m}) s_1(t_{21}) \dots s_m(t_{2m}) \dots s_1(t_{n1}) \dots s_m(t_{nm}), \quad (34)$$

the membrane potential will be at least v^+ or the neuron will have fired.

$$v(t_{nm}^+) \geq v^+ \vee \exists 1 \leq \nu \leq n, \exists 1 \leq \mu \leq m, v(t_{\nu\mu}^+) \geq \bar{v} \quad (35)$$

Proof. If $v(0) \geq v^-$, let $n = 1$, by Lemma 2, after applying the sequence, either the neuron has fired, or there exists $f \in \mathcal{R}_{\underline{v}}$, and by (29),

$$v(t_{1m}^+) = f(v(0)) \geq f(v^-) = v^+ \quad (36)$$

If $v(0) < v^-$, after applying the sequence, the neuron either has fired, or there exists $f \in \mathcal{R}_{\underline{v}}$, and by (30), $v(t_{1m}^+) = f(v(0)) \geq v(0) + f(v^-) = v^+ = v(0) + v^+ - v^-$. Using mathematical induction, we can prove that $\forall \nu \in \mathbb{N}^*$, either the neuron has fired or

$$v(t_{\nu m}^+) \geq v(0) + \nu(v^+ - v^-). \quad (37)$$

Let

$$n = \left\lceil \frac{v^- - v(0)}{v^+ - v^-} \right\rceil + 1 \quad (38)$$

where $\lceil \cdot \rceil$ represents the ceiling function. After the $(n-1)$ -th repeat of the sequence, either the neuron has fired, or $v(t_{(n-1)m}^+) \geq v^-$, which, similar to (36), results in that either the neuron fires or $v(t_{nm}^+) \geq v^+$. $\square$

Theorem 5. *For the network with only two layer, it is impossible to encode for the sequence of $M \geq 3$.*

Proof. For simplicity, we only prove the case where $M = 3$.

Based on the weights $W_0^{1,c}$, we can divide the neurons into the following categories

$$\mathcal{X}_{(u_{x-2}, u_{x-1}, u_x) \in \{-, 0, +\}^3} = \left\{ (c, y) : \bigwedge_{\xi=x-2}^x \text{sign}(w_0^{1,c}(y - \xi)) = u_\xi \right\} \quad (39)$$

When applying input s_{x-2}^0 repeatedly,

$\square$

4.1.2 Infeasible without multicasting

Theorem 6. *For a network with three layers, while the first two layers are defined as in 3.2, if only connections between adjacent layers are allowed, i.e. $\forall k \leq 1, W_k^{2,0} \equiv W_k^{2,1} \equiv \mathbf{0}$, it is infeasible to encode for the sequence of three.*

Proof. Consider input spike sequences $x, x+1, x+2$, the output spike sequence of layer 2 is $x, x+1$ of channel 0. For input sequence $x+1, x+2, x, x+1$, the output spike sequence of layer 2 is $x+1, x$. Thus if neuron x of channel 0 on layer 3 should respond to the former but not the latter (by problem definition), it will have to encode sequence of two from layer 2, which is impossible as proven in Sec. 3.1. $\square$

4.2 Solution with three layers allowing multicasting

The first two layers use the same configurations as defined in 3.2.

For layer 3, $C_3 = 2$ (by problem definition), $\underline{v}_3 = -1$, $\bar{v}_3 = 2$, $J_0^3 = 3$, $J_1^3 = 0$, $J_2^3 = 2$.

$$W_0^{3,0} = [\dots \quad -1 \quad -1 \quad -1 \quad 0 \quad 0 \quad -2 \quad -1 \quad \dots], \quad (40)$$

$$W_0^{3,1} = [\dots \quad -1 \quad -2 \quad 0 \quad 0 \quad -1 \quad -1 \quad -1 \quad \dots], \quad (41)$$

$$W_1^{3,0} = W_1^{3,1} = \mathbf{0} \quad (42)$$

$$W_2^{3,0} = \begin{bmatrix} \dots & 0 & 0 & 1 & 2 & 0 & \dots \\ \dots & 0 & 0 & 0 & -1 & 0 & \dots \end{bmatrix}, \quad (43)$$

$$W_2^{3,1} = \begin{bmatrix} \dots & 0 & -1 & 0 & 0 & 0 & \dots \\ \dots & 0 & 2 & 1 & 0 & 0 & \dots \end{bmatrix}, \quad (44)$$

The transition dynamics is shown in Fig. 4. By treating state “100,-1” as s_1 , state “010,1” as s_2 and all the others as s_0 , the state transition is equivalent to Fig. 1 for $M = 3$. To illustrate the working principle, the following cases are considered

- At any time when s_{x-2}^0 comes, $v_x^{3,1}$ will be reset to

$$\max \left(v_x^{3,1} + w_{1,1}^{3,1}(2), v_3 \right) = \max \left(\underbrace{v_x^{3,1}}_{\leq 1} - 2, -1 \right) = -1 \quad (45)$$

- For the encoded sequence $(x-2)^+(x-1)^+x$, the spikes from the layer 0 and 2 are

$$\underbrace{s_{x-2}^0}_{\times \geq 1} s_{x-1}^0 s_{x-1}^{2,1} \underbrace{s_{x-1}^0}_{\times \geq 0} s_x^0 s_x^{2,1} \underbrace{s_x^0}_{\times \geq 0} \quad (46)$$

and $v_x^{2,1}$ will undergo

$$\underbrace{(-2)_{\text{clip to } -1}}_{\times \geq 1} + 0 + 2 \underbrace{+0}_{\times \geq 0} + 0 + 1 \underbrace{+0}_{\times \geq 0} \quad (47)$$

and will fire exactly once after the first s_x^0 , which proves that the neuron will not miss any encoded sequence.

- The output neuron could not be triggered by $s_{x-1}^{2,1}$'s, as it will always be preceded by s_{x-2}^0 , which will reset the neuron as shown above. This is similar to Sec.3.2 where s_{x-1}^0 cannot trigger output $s_x^{2,1}$.
- The output neuron could not be triggered solely by repeating $s_x^{2,1}$'s, because after the first $s_x^{2,1}$, say at time t_1 , we have $v_x^{3,1}(t_1^+) = 1$. in order to get the next $s_x^{2,1}$, there must be at least one more input s_{x-1}^0 at, say $t_2(t_2 > t_1)$. Consider the input spikes during (t_1, t_2) :

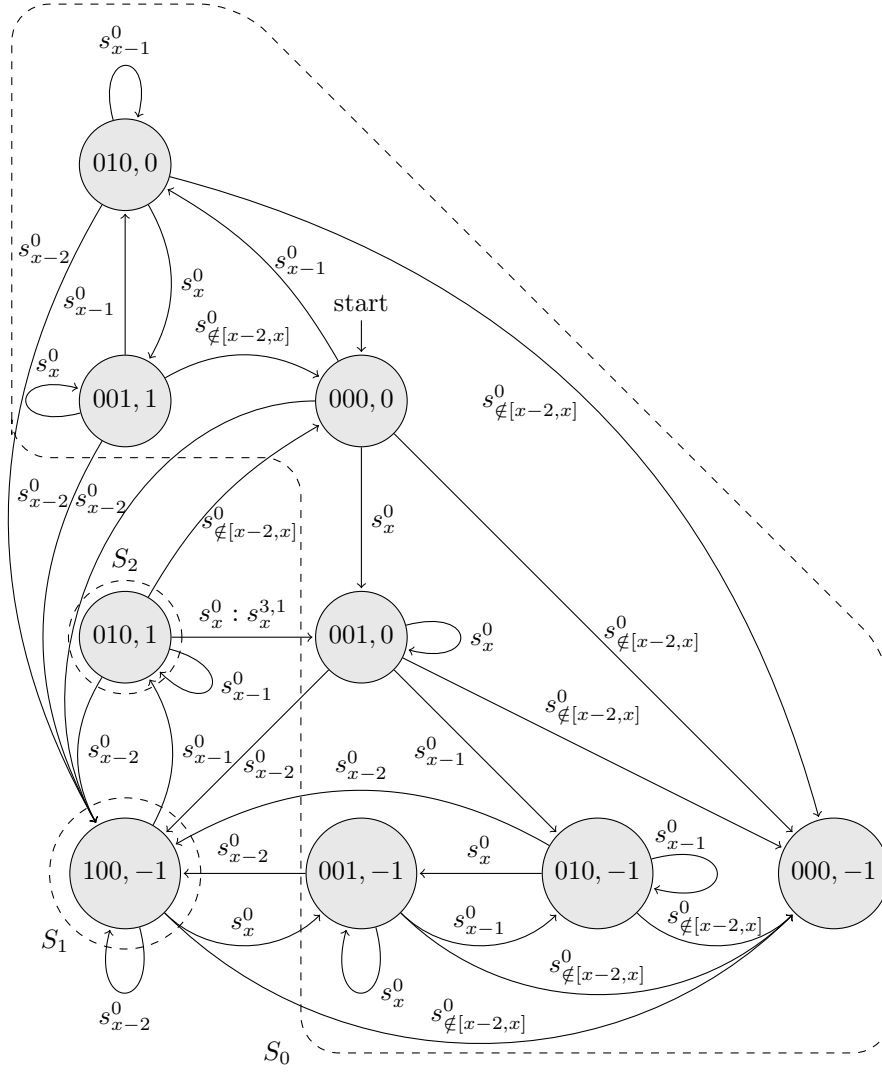


Figure 4: Transition dynamics for detecting sequence of three, the value in the circle represents $\left(\left[v_{x-1}^{2,1} \right]_+ \left[v_x^{2,1} \right]_+ \left[v_{x-1}^{2,2} \right]_+, v_x^{3,1} \right)$, whereas $[\cdot]_+ = \max(\cdot, 0)$ equivalently represents state $S_{[\cdot]_+}$

- If there is nothing or only repeating s_{x-1}^0 's, $s_{x-1}^{2,2}$ will fire shortly after t_2 , resetting $v_x^{3,1}(t_2^+) = v_x^{3,1}(t_2^-) + w_{2,2}^{3,1}(1) = 0$;
- If there is a s_{x-2}^0 in between, $v_x^{3,1}$ will be directly reset to -1;
- If there is any $s_{y \notin [x-2, x]}^0(t_y)$, $v_x^{3,1}(t_y^+) = v_x^{3,1}(t_y^-) + w_{0,1}^{3,1}(x-y) = 0$.

5 Sequence of ≥ 4 inputs with four layers

Inspired by the solution in Sec. 4.2, would it be possible to extend the sequence length to $M \geq 3$ with the same working principle?

5.1 Infeasible with three layers

Theorem 7. *For a network with three layers, while the first two layers are defined as in 3.2, it is infeasible to encode for the sequence of $M \geq 4$.*

Proof. Consider input sequence

$$\underbrace{s_{x-3}^0}_{\times \geq 1} \underbrace{s_{x-2}^0}_{\times \geq 1} \underbrace{s_{x-1}^0}_{\times \geq 1} \underbrace{s_x^0}_{\times \geq 1} \quad (48)$$

including spikes of layer 2,

$$\underbrace{s_{x-3}^0}_{\times \geq 1} s_{x-2}^0 s_{x-2}^{2,1} \underbrace{s_{x-2}^0}_{\times \geq 0} s_{x-1}^0 s_{x-1}^{2,1} \underbrace{s_{x-1}^0}_{\times \geq 0} s_x^{0,2,1} \underbrace{s_x^0}_{\times \geq 0} \quad (49)$$

since $s_{x-2}^0, s_{x-1}^0, s_x^0$ could repeat infinite many times, the corresponding weights must all be zero

$$w_{0,1}^{3,1}(0) = w_{0,1}^{3,1}(1) = w_{0,1}^{3,1}(2) = 0 \quad (50)$$

Thus to consider only incoming spikes with nonzero weights, (49) can be simplified to

$$\underbrace{s_{x-3}^0}_{\times \geq 1} s_{x-2}^{2,1} s_{x-1}^{2,1} s_x^{2,1} \quad (51)$$

The neuron shouldn't fire after $s_{x-2}^{2,1}$, let V be the membrane potential at this moment, we have

$$\underline{v} \leq V < \bar{v} \quad (52)$$

The after $s_{x-1}^{2,1}$ the neuron still doesn't fire, thus

$$\max(V + w_{2,1}^{3,1}(1), \underline{v}) < \bar{v} \quad (53)$$

But after $s_x^{2,1}$ the neuron fires, thus

$$\max(V + w_{2,1}^{3,1}(1), \underline{v}) + w_{2,1}^{3,1}(0) \geq \bar{v} \quad (54)$$

The l.h.s., defined as function

$$f(v) := \max \left(v + w_{2,1}^{3,1}(1, \underline{v}) + w_{2,1}^{3,1}(0) \right) \quad (55)$$

describes the dynamics of the membrane potential $v_x^{3,1}$ when applying input sequence

$$s_{x-2}^0 s_{x-1}^0 s_x^0 \quad (56)$$

whose output spikes (with non-zero weights) at layer 2 are

$$s_{x-1}^{2,1} s_x^{2,1} \quad (57)$$

It is easy to show that

$$f(v) \geq f(V) + \min(v - V, 0) \quad (58)$$

Thus,

- If $v \leq V$,

$$f(v) - v \geq f(V) - V \quad (59)$$

means that applying sequence (56) will increase the membrane potential by at least a positive constant value $f(V) - V > 0$.

- If $v > V$,

$$f(v) - v \geq f(V) \geq \bar{v} \quad (60)$$

means that applying sequence (56) will make the neuron fire.

Both contradicts with the problem definition that repeating sequence

$$\underbrace{s_{x-2}^0 s_{x-1}^0 s_x^0}_{\times \geq 1} \quad (61)$$

should not make the neuron fire. $\square$

5.2 Solution for sequences of length $M \geq 4$ with four layers

The first three layers use the same configurations as defined in 4.2.

For layer 4, $C_4 = 2$ (by problem definition), $\underline{v}_4 = -1$, $\bar{v}_4 = 2$, $J_0^4 = J_1^4 = 0$, $J_2^4 = J_3^4 = M - 2$.

$$W_0^{4,1} = W_0^{4,2} = W_1^{4,1} = W_1^{4,2} = \mathbf{0} \quad (62)$$

$$W_2^{4,1} = \begin{bmatrix} \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & -1 & \cdots & -1 & -2 & 0 & \cdots \\ \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \end{bmatrix} \quad (63)$$

$\underbrace{\hspace{1.5cm}}_{\times(M-4)} \quad \underbrace{\hspace{1.5cm}}_{\times(M-4)}$

$$W_2^{4,2} = \begin{bmatrix} \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \\ \cdots & 0 & -2 & -1 & \cdots & -1 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \end{bmatrix} \quad (64)$$

$\underbrace{\hspace{1.5cm}}_{\times(M-4)} \quad \underbrace{\hspace{1.5cm}}_{\times(M-4)}$

$$W_3^{4,1} = \begin{bmatrix} \cdots & 0 & 0 & 0 & \cdots & 0 & 1 & 1 & \cdots & 1 & 2 & 0 & \cdots \\ \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \end{bmatrix} \quad (65)$$

$\underbrace{\hspace{10em}}_{\times(M-4)} \qquad \underbrace{\hspace{10em}}_{\times(M-4)}$

$$W_3^{4,2} = \begin{bmatrix} \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots \\ \cdots & 0 & 2 & 1 & \cdots & 1 & 1 & 0 & \cdots & 0 & 0 & 0 & \cdots \end{bmatrix} \quad (66)$$

$\underbrace{\hspace{10em}}_{\times(M-4)} \qquad \underbrace{\hspace{10em}}_{\times(M-4)}$

Proof. It can be shown that the transitions of the tuple consists of equivalent states of neurons in layer 2 and 3 defined in Fig. 3 and Fig. 4 and the membrane potential of $v_x^{4,1}$

$$T := \left(S_{x-M+3}^{3,1} S_{x-M+3}^{2,1} \cdots, S_{x-1}^{3,1} S_{x-1}^{2,1} S_x^{3,1}, v_x^{4,1} \right) \quad (67)$$

is equivalent to the finite state machine defined in Fig. 1, i.e.

$$\left\{ \begin{array}{ll} \mathcal{T}_0 := \bigcup_{v \in [-1,1]} \bigcup_{i=1}^{M-3} \left\{ \underbrace{(0 \cdots 0}_{2i-1} 110 \cdots 0, v) \right\} \cup \{(0, v)\} & \rightarrow S_0 \\ \mathcal{T}_1 := \bigcup_{v \in [-1,1]} \{(10 \cdots 0, v)\} & \rightarrow S_1 \\ \mathcal{T}_2 := \bigcup_{v \in [-1,1]} \{(2110 \cdots 0, v)\} & \rightarrow S_2 \\ \forall m \in [3, M-2], \mathcal{T}_m := \left\{ \underbrace{(0 \cdots 0}_{2m-4} 2110 \cdots 0, 1) \right\} & \rightarrow S_m \\ \mathcal{T}_{M-1} := \{(0 \cdots 02, 1)\} & \rightarrow S_{M-1} \end{array} \right. \quad (68)$$

□